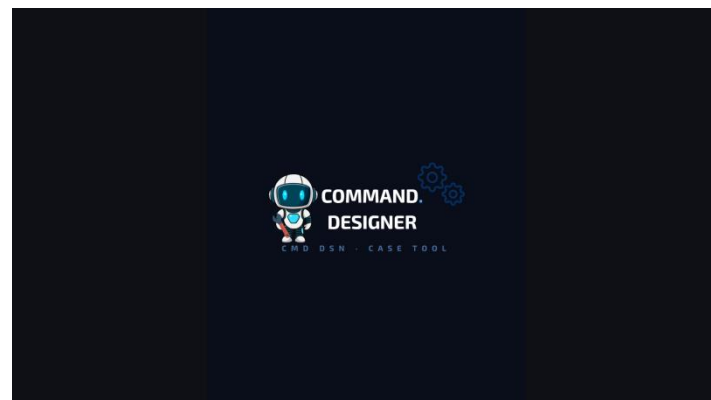
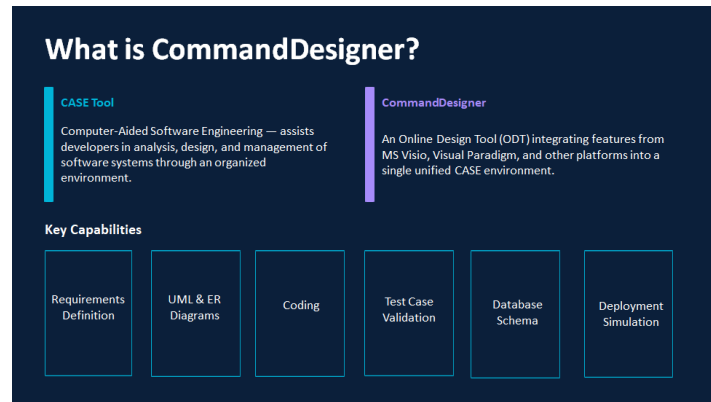


Week 03 :Curate Your Images

Raaeel Fatima

CommandDesigner



InventoryIQ:

```
cout << "\n-----\n";
cout << "      OPTIMAL INVENTORY DECISION      \n";
cout << "-----\n";

int totalProfit = 0;
int totalWastage = 0;

for (const auto& p : products) {
    int sellQty = min(p.stock, p.demand);
    int profit = calculateProfit(p, sellQty);
    int wastage = max(0, p.stock - p.demand);
    int reorder = max(0, p.demand - p.stock);

    totalProfit += profit;
    totalWastage += wastage;

    cout << "\nProduct      : " << p.name << "\n";
    cout << "Profit Margin : " << (p.sellPrice - p.costPrice) << " per unit\n";
    cout << "Sell Quantity : " << sellQty << "\n";
    cout << "Profit Earned : " << profit << "\n";
    cout << "Wastage       : " << wastage << " units\n";

    int profitMargin = p.sellPrice - p.costPrice;
    bool isProfitable = (profitMargin > 0);
    int nextOrder = 0;

    if (isProfitable) {
        int safetyStock = p.demand * 0.2;
        nextOrder = p.demand + safetyStock;
    }
}
```

E-Commerce:

```
int main() {
    AVLNode* root = nullptr;
    loadProducts(root);
    loadUsers();
    loadOrders();
    loadRelations();

    cout << "-----\n REGISTER ADMIN FIRST \n-----\n";
    addUser();

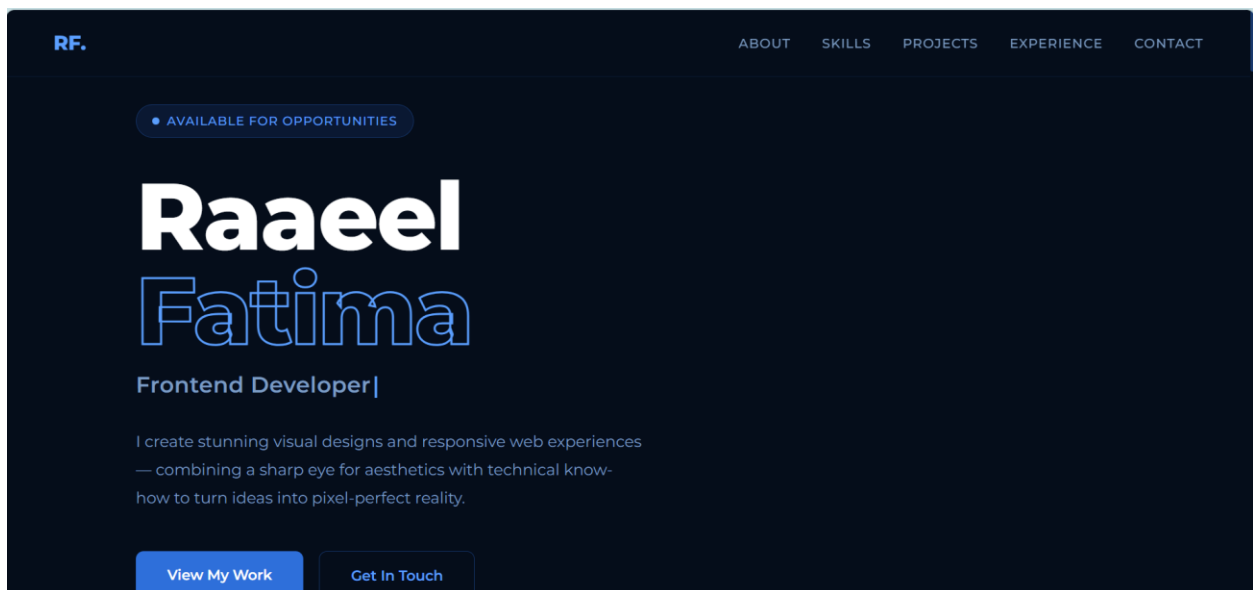
    cout << "\n----- LOGIN ----- \n";
    if (!loginUser(currentUser)) { cout << "Invalid login! Exiting.\n"; return 0; }
    cout << "\nLogin Successful!\n";

    loadCart();

    int choice;
    do {
        cout << "\n===== E-COMMERCE MENU =====\n";
        cout << "1. Add Product (AVL)\n2. View All Products\n3. Search Product\n4. Add to Cart\n5. View Cart\n6. Place Order (Priority)\n7. \n";
        cin >> choice;

        switch (choice) {
case 1: {
            Product p;
            cout << "Enter Product ID: "; cin >> p.id;
            cout << "Enter Name: "; cin >> p.name;
            cout << "Enter Price: "; cin >> p.price;
```

Portfolio:



```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Raheel Fatima - Portfolio</title>
7  <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@300;400;500;600;700;800&display=swap" rel="stylesheet">
8  <style>
9  img{max-width:100%;display:block}
10 /*{margin:0;padding:0;box-sizing:border-box}
11 :root{
12   --navy: #050d1a;--navy2: #0a1628;--navy3: #0f2040;
13   --blue: #1a3a6e;--accent: #2e6fda;--accent2: #5b9fff;
14   --glow: #3a7bd5;--text: #e8f0fe;--muted: #7b9cc4;
15   --border: #1e3a6b;--card: #0d1f3c;
16 }
17 html{scroll-behavior:smooth}
18 body{font-family: 'Montserrat', sans-serif;background:var(--navy);color:var(--text);overflow-x:hidden;line-height:1.7}
19
20 /* NAV */
21 nav{position:fixed;top:0;left:0;right:0;z-index:100;display:flex;align-items:center;justify-content:space-between;padding:1rem 3rem;background:var(--navy);border-bottom:1px solid var(--border)}
22 nav.scrolled{box-shadow:0 2px 30px #000000;background:var(--navy2)}
23 .logo{font-weight:800;font-size:1.3rem;color:var(--accent2);letter-spacing:-0.5px}
24 .nav-links{display:flex;gap:2rem;list-style:none}
25 .nav-links a{font-size:0.85rem;font-weight:500;color:var(--muted);text-decoration:none;letter-spacing:0.04em;text-transform:uppercase;transition:color 0.3s}
26 .nav-links a::after{content:'';position:absolute;bottom:-3px;left:0;right:0;height:2px;background:var(--accent2);transform:scaleX(0);transition:transform 0.3s}
27 .nav-links a:hover{color:var(--accent2)}
28 .nav-links a:hover::after{transform:scaleX(1)}
29

```

Mini Os Kernal:

```

    int turnaround;
};

// ===== FCFS SCHEDULING =====
void FCFS(vector<Process> processes) {
    int time = 0;

    cout << "\n--- FCFS Scheduling ---\n";

    for (auto &p : processes) {
        if (time < p.arrival)
            time = p.arrival;

        time += p.burst;
        p.completion = time;
        p.turnaround = p.completion - p.arrival;
        p.waiting = p.turnaround - p.burst;

        cout << "P" << p.pid << " WT: " << p.waiting
              << " TAT: " << p.turnaround << endl;
    }
}

// ===== ROUND ROBIN =====
void RoundRobin(vector<Process> processes, int quantum) {
    queue<int> q;
    int time = 0;

```

Problem Statement

Design and implement a Mini OS Kernel Simulator in user space that models fundamental OS functionalities including process management, CPU scheduling, synchronization, and memory management with configurable policies and quantitative performance evaluation.



Process & Thread
Management



CPU Scheduling
Algorithms



Process
Synchronization



Memory
Management

Tools & Language

C++ (Core Kernel)

HTML/CSS/JS (GUI)

Dev C++ / MinGW GCC

draw.io (Diagrams)

Microsoft Edge (to run
the GUI)

Identity kit:

RAAEEL FATIMA IDENTITY KIT

TYPOGRAPHY

Heading font information

Syne
Headings / Titles

Body font

DM Sans
Body / Paragraphs / UI text

Aa — Syne

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
01 02 03 04

a — DM Sans

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
01 02 03 04

COLOR PALETTE



LOGO / FAVICON

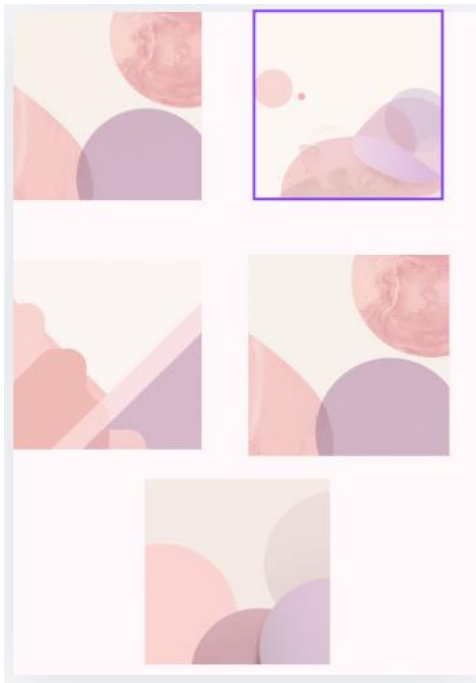


Fonts: Syne for headings and DM Sans for body text;
palette: #D96C9D, #1A1A1A, #FFF9FC, #8E6BBE.
Mood: Soft, feminine, modern, and minimal, keeping
the focus on the work rather than the visual style.

AI Generated Images:



Selected because its soft pink, purple, and off-white palette matches my identity kit and its minimal composition keeps attention on the actual portfolio work.



Rejected: I rejected the other generated options because they felt more decorative and visually busy. Their stronger shapes and compositions could compete with my actual project screenshots, so I selected the quieter option to keep the focus on my work.

Real vs AI decision

Last section:

Real vs AI decisions: I used real screenshots for my projects because they provide authentic evidence of my work and implementation. I chose not to use AI-generated stand-ins for project visuals. For my personal identity, I used my RF monogram instead of a personal photograph due to privacy considerations. AI was used only for a subtle decorative/hero visual that complements the identity kit.